Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

Friday 1 May 2026

1.1 Daily Limit

Logarithmic Limit of an Infinite Cosine Product

Evaluate the limit:

$$\lim_{n\to\infty}\sum_{i=1}^{n}\ln\left(\cos\left(\frac{\frac{\pi}{2}}{2^i}\right)\right)$$

Solution: Using the fundamental properties of logarithms, we can convert the finite sum of logarithmic terms into the natural logarithm of a finite product:

$$S_n = \sum_{i=1}^{n}\ln\left(\cos\left(\frac{\pi}{2^{i+1}}\right)\right) = \ln\left(\prod_{i=1}^{n}\cos\left(\frac{\pi}{2^{i+1}}\right)\right)$$

Let us focus on evaluating the inner product, which we denote as P_n:

$$P_n = \cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{8}\right)\cdots\cos\left(\frac{\pi}{2^{n+1}}\right)$$

We can evaluate this product elegantly using the sine double-angle identity, $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$, which can be rearranged as $\cos(\theta) = \frac{\sin(2\theta)}{2\sin(\theta)}$. Multiply and divide the entire product P_n by $2^n\sin\left(\frac{\pi}{2^{n+1}}\right)$:

$$P_n = \frac{2^n\sin\left(\frac{\pi}{2^{n+1}}\right)\cos\left(\frac{\pi}{2^{n+1}}\right)\cos\left(\frac{\pi}{2^n}\right)\cdots\cos\left(\frac{\pi}{4}\right)}{2^n\sin\left(\frac{\pi}{2^{n+1}}\right)}$$

Notice how the terms in the numerator sequentially telescope from right to left: 1. Combining the first pair: $2\sin\left(\frac{\pi}{2^{n+1}}\right)\cos\left(\frac{\pi}{2^{n+1}}\right) = \sin\left(\frac{\pi}{2^n}\right)$. 2. Combining with the next cosine term: $2\sin\left(\frac{\pi}{2^n}\right)\cos\left(\frac{\pi}{2^n}\right) = \sin\left(\frac{\pi}{2^{n-1}}\right)$.

Continuing this cascading chain of double-angle simplifications across all n terms reduces the entire numerator to a single sine function evaluated at double the largest angle:

$$P_n = \frac{\sin\left(2\cdot\frac{\pi}{4}\right)}{2^n\sin\left(\frac{\pi}{2^{n+1}}\right)} = \frac{\sin\left(\frac{\pi}{2}\right)}{2^n\sin\left(\frac{\pi}{2^{n+1}}\right)} = \frac{1}{2^n\sin\left(\frac{\pi}{2^{n+1}}\right)}$$

We now take the limit of this product as $n\to\infty$. To evaluate the denominator cleanly, we introduce the substitution $\theta_n = \frac{\pi}{2^{n+1}}$. Notice that as $n\to\infty$, $\theta_n\to 0^+$, and we can rewrite the factor 2^n as $\frac{\pi}{2\theta_n}$:

$$\lim_{n\to\infty}P_n = \lim_{\theta_n\to 0}\frac{1}{\frac{\pi}{2\theta_n}\sin(\theta_n)} = \frac{2}{\pi}\lim_{\theta_n\to 0}\frac{\theta_n}{\sin(\theta_n)}$$

Using the standard trigonometric limit $\lim_{\theta\to 0}\frac{\theta}{\sin(\theta)} = 1$, we obtain:

$$\lim_{n\to\infty}P_n = \frac{2}{\pi}\cdot 1 = \frac{2}{\pi}$$

Substituting this simplified product limit back into our logarithmic expression yields the final exact value:

$$L = \ln\left(\lim_{n\to\infty}P_n\right) = \ln\left(\frac{2}{\pi}\right) = \ln(2) - \ln(\pi)$$

Saturday 2 May 2026

2.1 Daily Limit

Infinite Product Involving Cubic Expressions

Evaluate the limit:

$$\lim_{n\to\infty}\prod_{k=2}^{n}\frac{k^3-1}{k^3+1}$$

Solution: We begin by algebraically factoring both the numerator and the denominator using the difference and sum of cubes formulas:

$$k^3-1=(k-1)(k^2+k+1)$$

$$k^3+1=(k+1)(k^2-k+1)$$

Notice the algebraic relationship between the quadratic factors: if we define $g(k)=k^2-k+1$, evaluating $g(k+1)$ yields:

$$g(k+1)=(k+1)^2-(k+1)+1=k^2+2k+1-k-1+1=k^2+k+1$$

This allows us to split the general product into two distinct telescoping products:

$$\prod_{k=2}^{n}\frac{k^3-1}{k^3+1}=\left(\prod_{k=2}^{n}\frac{k-1}{k+1}\right)\left(\prod_{k=2}^{n}\frac{g(k+1)}{g(k)}\right)$$

We evaluate the first rational product by writing out the initial and final terms to observe the systematic cancellation:

$$\prod_{k=2}^{n}\frac{k-1}{k+1}=\frac{1}{3}\cdot\frac{2}{4}\cdot\frac{3}{5}\cdots\frac{n-2}{n}\cdot\frac{n-1}{n+1}=\frac{1\cdot 2}{n(n+1)}=\frac{2}{n(n+1)}$$

Next, we evaluate the second quadratic product, which telescopes directly:

$$\prod_{k=2}^{n}\frac{g(k+1)}{g(k)}=\frac{g(3)}{g(2)}\cdot\frac{g(4)}{g(3)}\cdots\frac{g(n+1)}{g(n)}=\frac{g(n+1)}{g(2)}$$

Substituting $g(2)=2^2-2+1=3$ and $g(n+1)=n^2+n+1$:

$$\prod_{k=2}^{n}\frac{g(k+1)}{g(k)}=\frac{n^2+n+1}{3}$$

Multiplying both simplified products together gives the exact expression for the n-th partial product P_n:

$$P_n=\frac{2}{n(n+1)}\cdot\frac{n^2+n+1}{3}=\frac{2(n^2+n+1)}{3(n^2+n)}$$

Taking the limit as $n\to\infty$, the dominant quadratic terms in the numerator and denominator govern the asymptotic behavior:

$$\lim_{n\to\infty}P_n=\lim_{n\to\infty}\frac{2n^2+2n+2}{3n^2+3n}=\frac{2}{3}$$

Sunday 3 May 2026

3.1 Daily Limit

Limit of an Infinite Series of Iterated Integrals

Evaluate the limit:

$$\lim_{x\to 0}\frac{1}{x^2}\sum_{m=1}^{\infty}\int_0^x\int_0^{t_1}\cdots\int_0^{t_{m-1}}\ln(1+t_m)\,dt_m\cdots dt_1$$

Solution: Let $I_m(x)$ denote the m-fold iterated integral of the function $f(t)=\ln(1+t)$ over the interval $[0,x]$. By Cauchy's Formula for Repeated Integration, an m-fold nested integral can be reduced to a single definite integral:

$$I_m(x)=\frac{1}{(m-1)!}\int_0^x(x-t)^{m-1}\ln(1+t)\,dt$$

Summing this expression over all positive integers m from 1 to ∞:

$$S(x)=\sum_{m=1}^{\infty}I_m(x)=\sum_{m=1}^{\infty}\int_0^x\frac{(x-t)^{m-1}}{(m-1)!}\ln(1+t)\,dt$$

Since the series converges uniformly on the domain $[0,x]$, we can interchange the order of summation and integration. Using the index shift $k=m-1$:

$$S(x)=\int_0^x\left(\sum_{k=0}^{\infty}\frac{(x-t)^k}{k!}\right)\ln(1+t)\,dt$$

We recognize the inner summation as the Maclaurin series expansion of the exponential function e^{x-t}:

$$S(x)=\int_0^x e^{x-t}\ln(1+t)\,dt=e^x\int_0^x e^{-t}\ln(1+t)\,dt$$

We are tasked with evaluating the limit:

$$L=\lim_{x\to 0}\frac{S(x)}{x^2}=\lim_{x\to 0}\frac{\int_0^x e^{x-t}\ln(1+t)\,dt}{x^2}$$

As $x\to 0$, both the numerator $S(0)=0$ and the denominator $0^2=0$ vanish, presenting an indeterminate form of type $\frac{0}{0}$. We apply L'Hôpital's Rule by differentiating numerator and denominator with respect to x. Using the Fundamental Theorem of Calculus along with the Leibniz Integral Rule:

$$S'(x)=\frac{d}{dx}\left[e^x\int_0^x e^{-t}\ln(1+t)\,dt\right]=e^x\left(e^{-x}\ln(1+x)\right)+e^x\int_0^x e^{-t}\ln(1+t)\,dt=\ln(1+x)+S(x)$$

Applying L'Hôpital's Rule yields:

$$L=\lim_{x\to 0}\frac{S'(x)}{2x}=\lim_{x\to 0}\frac{\ln(1+x)+S(x)}{2x}=\frac{1}{2}\lim_{x\to 0}\frac{\ln(1+x)}{x}+\frac{1}{2}\lim_{x\to 0}\frac{S(x)}{x}$$

The first standard limit evaluates identically to 1:

$$\lim_{x\to 0}\frac{\ln(1+x)}{x}=1$$

For the second term, since $S(x) = O(x^2)$ as $x \to 0$, we have $\lim_{x\to 0} \frac{S(x)}{x} = 0$. Substituting these values back into our limit expression:

$$L = \frac{1}{2}(1) + \frac{1}{2}(0) = \frac{1}{2}$$

Monday 4 May 2026

4.1 Daily Limit

Limit of an n-th Root of a Permutation Product

Evaluate the limit:

$$\lim_{n\to\infty} \frac{1}{n} \sqrt[n]{(n+1)(n+2)\cdots(2n)}$$

Solution: Let L denote the limit. We can bring the factor $\frac{1}{n} = \sqrt[n]{\frac{1}{n^n}}$ directly inside the n-th root radical:

$$L = \lim_{n\to\infty} \left[\frac{(n+1)(n+2)\cdots(n+n)}{n^n} \right]^{\frac{1}{n}} = \lim_{n\to\infty} \left[\prod_{k=1}^{n} \left(\frac{n+k}{n} \right) \right]^{\frac{1}{n}}$$

Simplifying each fraction within the product:

$$L = \lim_{n\to\infty} \left[\prod_{k=1}^{n} \left(1 + \frac{k}{n} \right) \right]^{\frac{1}{n}}$$

To convert this multiplicative expression into an additive sum, we take the natural logarithm of both sides:

$$\ln L = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ln\left(1 + \frac{k}{n} \right)$$

We recognize this expression as a right-endpoint Riemann sum for the continuous function $f(x) = \ln(1+x)$ over the interval $[0,1]$, where $\Delta x = \frac{1}{n}$ and $x_k = \frac{k}{n}$. Passing to the definite integral:

$$\ln L = \int_0^1 \ln(1+x)\, dx$$

We evaluate this integral using integration by parts. Let $u = \ln(1+x)$ and $dv = dx$, which gives $du = \frac{1}{1+x} dx$ and $v = 1+x$:

$$\int_0^1 \ln(1+x)\, dx = \left[(1+x)\ln(1+x) - x \right]_0^1$$

Evaluating at the upper and lower bounds:

$$\ln L = (2\ln(2) - 1) - (1\ln(1) - 0) = \ln(4) - 1 = \ln(4) - \ln(e) = \ln\left(\frac{4}{e} \right)$$

Exponentiating both sides to solve for L yields the exact value:

$$L = \frac{4}{e}$$

Tuesday 5 May 2026

5.1 Daily Limit

Asymptotic Limit of a Sine Power Integral

Evaluate the limit:

$$\lim_{n\to\infty} \sqrt{n} \int_0^{\pi} \sin^n(x)\, dx$$

Solution: Let $I_n = \int_0^{\pi} \sin^n(x)\, dx$. Notice that the integrand $\sin^n(x)$ is symmetric about $x = \frac{\pi}{2}$ on the interval $[0, \pi]$. Therefore, we can express the integral as:

$$I_n = 2\int_0^{\frac{\pi}{2}} \sin^n(x)\, dx$$

As the exponent $n \to \infty$, the function $\sin^n(x)$ becomes sharply peaked around its maximum value at $x = \frac{\pi}{2}$, while exponentially vanishing everywhere else. To apply Laplace's Method, we introduce the local variable substitution $x = \frac{\pi}{2} - t$. Then $dx = -dt$, and the integral transforms to:

$$I_n = 2\int_0^{\frac{\pi}{2}} \cos^n(t)\, dt$$

For small values of t, we use the Taylor series expansion of the cosine function:

$$\cos(t) = 1 - \frac{t^2}{2} + O(t^4)$$

Using the exponential limit identity $(1-u) \approx e^{-u}$ for small u, we approximate the integrand:

$$\cos^n(t) = \left(1 - \frac{t^2}{2} + O(t^4)\right)^n \approx \left(e^{-\frac{t^2}{2}}\right)^n = e^{-\frac{nt^2}{2}}$$

Since the primary contribution to the integral occurs in the immediate vicinity of $t = 0$, we substitute this Gaussian approximation and safely extend the upper limit of integration to infinity without affecting the asymptotic behavior as $n \to \infty$:

$$I_n \sim 2\int_0^{\infty} e^{-\frac{nt^2}{2}}\, dt$$

We evaluate this Gaussian integral using the substitution $u = t\sqrt{\frac{n}{2}}$, which implies $dt = \sqrt{\frac{2}{n}}\, du$:

$$I_n \sim 2\sqrt{\frac{2}{n}} \int_0^{\infty} e^{-u^2}\, du$$

Recalling the standard Gaussian probability integral $\int_0^{\infty} e^{-u^2}\, du = \frac{\sqrt{\pi}}{2}$:

$$I_n \sim 2\sqrt{\frac{2}{n}} \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\frac{2\pi}{n}}$$

Substituting this asymptotic behavior back into the original limit expression:

$$L = \lim_{n\to\infty} \sqrt{n} I_n = \lim_{n\to\infty} \sqrt{n} \cdot \sqrt{\frac{2\pi}{n}} = \sqrt{2\pi}$$

Wednesday 6 May 2026

6.1 Daily Limit

Limit of a Difference of Exponential Fractions

Evaluate the limit:

$$\lim_{x\to\infty}\left(\frac{x^x}{(x-1)^{x-1}}-\frac{(x-1)^{x-1}}{(x-2)^{x-2}}\right)$$

Solution: Let us define the generating function $f(x)=\frac{x^x}{(x-1)^{x-1}}$. The required limit is simply the asymptotic difference $\lim_{x\to\infty}[f(x)-f(x-1)]$. We begin by algebraically rewriting $f(x)$:

$$f(x)=x\cdot\frac{x^{x-1}}{(x-1)^{x-1}}=x\left(\frac{x}{x-1}\right)^{x-1}=x\left(1-\frac{1}{x}\right)^{-(x-1)}$$

To determine the asymptotic expansion of this function for large x, we take the natural logarithm of the exponential term:

$$\ln\left[\left(1-\frac{1}{x}\right)^{-(x-1)}\right]=-(x-1)\ln\left(1-\frac{1}{x}\right)$$

Using the Maclaurin series expansion $\ln(1-u)=-u-\frac{u^2}{2}-\frac{u^3}{3}-O(u^4)$ where $u=\frac{1}{x}$:

$$-(x-1)\left(-\frac{1}{x}-\frac{1}{2x^2}-\frac{1}{3x^3}-O(x^{-4})\right)=(x-1)\left(\frac{1}{x}+\frac{1}{2x^2}+\frac{1}{3x^3}+O(x^{-4})\right)$$

Expanding the product algebraically:

$$=1+\frac{1}{2x}+\frac{1}{3x^2}-\frac{1}{x}-\frac{1}{2x^2}+O(x^{-3})=1-\frac{1}{2x}-\frac{1}{6x^2}+O(x^{-3})$$

Exponentiating this expression using $e^{1+v}=e\cdot e^v=e\left(1+v+\frac{v^2}{2}+O(v^3)\right)$:

$$\left(1-\frac{1}{x}\right)^{-(x-1)}=e\cdot\exp\left(-\frac{1}{2x}-\frac{1}{6x^2}+O(x^{-3})\right)$$

$$=e\left[1+\left(-\frac{1}{2x}-\frac{1}{6x^2}\right)+\frac{1}{2}\left(-\frac{1}{2x}\right)^2+O(x^{-3})\right]=e\left[1-\frac{1}{2x}-\frac{1}{24x^2}+O(x^{-3})\right]$$

Multiplying through by the outer factor x yields the asymptotic expansion for $f(x)$:

$$f(x)=ex-\frac{e}{2}-\frac{e}{24x}+O(x^{-2})$$

Replacing x with $x-1$ gives the corresponding expansion for the shifted term $f(x-1)$:

$$f(x-1)=e(x-1)-\frac{e}{2}-\frac{e}{24(x-1)}+O(x^{-2})=ex-\frac{3e}{2}-\frac{e}{24x}+O(x^{-2})$$

Subtracting $f(x-1)$ from $f(x)$ cleanly cancels the divergent linear terms:

$$f(x)-f(x-1)=\left(ex-\frac{e}{2}\right)-\left(ex-\frac{3e}{2}\right)+O\left(\frac{1}{x}\right)=e+O\left(\frac{1}{x}\right)$$

Taking the limit as $x\to\infty$, all inverse power remainder terms vanish:

$$\lim_{x\to\infty}[f(x)-f(x-1)]=e$$

Thursday 7 May 2026

7.1 Daily Limit

Exponential Limit of a Nested Radical Involving 2

Evaluate the limit:

$$\lim_{n\to\infty} 2^n \sqrt{2 - \underbrace{\sqrt{2+\sqrt{2+\cdots+\sqrt{2}}}}_{n-1 \text{ radicals}}}$$

Solution: Let a_k denote the inner nested radical containing k square roots of 2. We evaluate the first few terms using trigonometric half-angle identities:

$$a_1 = \sqrt{2} = 2\cos\left(\frac{\pi}{4}\right)$$

$$a_2 = \sqrt{2+\sqrt{2}} = \sqrt{2+2\cos\left(\frac{\pi}{4}\right)} = \sqrt{4\cos^2\left(\frac{\pi}{8}\right)} = 2\cos\left(\frac{\pi}{8}\right)$$

By mathematical induction, for a nested radical containing $n-1$ square roots, we have the closed trigonometric form:

$$a_{n-1} = \underbrace{\sqrt{2+\sqrt{2+\cdots+\sqrt{2}}}}_{n-1 \text{ radicals}} = 2\cos\left(\frac{\pi}{2^n}\right)$$

Substituting this closed-form expression back into the limit:

$$L = \lim_{n\to\infty} 2^n \sqrt{2 - 2\cos\left(\frac{\pi}{2^n}\right)}$$

We simplify the expression under the square root using the fundamental half-angle identity $1-\cos(\theta) = 2\sin^2\left(\frac{\theta}{2}\right)$, setting $\theta = \frac{\pi}{2^n}$:

$$2 - 2\cos\left(\frac{\pi}{2^n}\right) = 2\left[1-\cos\left(\frac{\pi}{2^n}\right)\right] = 4\sin^2\left(\frac{\pi}{2^{n+1}}\right)$$

Taking the principal square root (since the angle $\frac{\pi}{2^{n+1}}$ lies strictly in the first quadrant for all $n \geq 1$, where sine is strictly positive):

$$\sqrt{2-2\cos\left(\frac{\pi}{2^n}\right)} = 2\sin\left(\frac{\pi}{2^{n+1}}\right)$$

Substituting this simplified radical back into our limit expression:

$$L = \lim_{n\to\infty} 2^n \cdot 2\sin\left(\frac{\pi}{2^{n+1}}\right) = \lim_{n\to\infty} 2^{n+1}\sin\left(\frac{\pi}{2^{n+1}}\right)$$

To evaluate this cleanly, we introduce the angular variable substitution $\theta_n = \frac{\pi}{2^{n+1}}$. Notice that as $n \to \infty$, $\theta_n \to 0^+$, and we can rewrite the coefficient as $2^{n+1} = \frac{\pi}{\theta_n}$:

$$L = \lim_{\theta_n\to 0} \frac{\pi}{\theta_n}\sin(\theta_n) = \pi \lim_{\theta_n\to 0}\frac{\sin(\theta_n)}{\theta_n}$$

Using the fundamental trigonometric limit $\lim_{\theta\to 0}\frac{\sin(\theta)}{\theta} = 1$:

$$L = \pi(1) = \pi$$

Friday 8 May 2026

8.1 Daily Limit

Improper Integral Involving a Composite Sine Exponential

Evaluate the integral:

$$\int_{-\infty}^{\infty} \sin(\pi e^x)\,dx$$

Solution: We evaluate this improper integral over the entire real line by applying a change of variables. Let us define the substitution:

$$u = \pi e^x$$

Differentiating both sides with respect to x gives the differential relationship:

$$du = \pi e^x\,dx = u\,dx \implies dx = \frac{du}{u}$$

We now determine the transformed boundary limits of integration: 1. As the lower bound $x \to -\infty$, the exponential function vanishes: $u = \pi e^{-\infty} \to 0^+$. 2. As the upper bound $x \to \infty$, the exponential grows without bound: $u = \pi e^{\infty} \to \infty$.

Substituting the new variable and integration limits into our definite integral:

$$I = \int_{-\infty}^{\infty} \sin(\pi e^x)\,dx = \int_{0}^{\infty} \sin(u) \cdot \frac{du}{u} = \int_{0}^{\infty} \frac{\sin(u)}{u}\,du$$

We recognize this transformed expression as the classical Dirichlet Integral. Its exact convergence value is a well-established result in analytical mathematics, frequently proven via Feynman's technique of differentiation under the integral sign or Laplace transforms:

$$\int_{0}^{\infty} \frac{\sin(u)}{u}\,du = \frac{\pi}{2}$$

Thus, the original improper integral converges strictly to:

$$\frac{\pi}{2}$$

Saturday 9 May 2026

9.1 Daily Limit

Limit of an Inverse Trigonometric Rational Expression

Evaluate the limit:

$$L = \lim_{x\to 0} \frac{\tan^{-1}\left(\frac{x^5}{1+\sqrt{1-x^{10}}}\right) - \frac{x^5}{2}}{x^{15}}$$

Solution: To simplify the algebraic argument of the inverse tangent function, we make the polynomial variable substitution $u = x^5$. Note that as $x \to 0$, we necessarily have $u \to 0$. The limit transforms to:

$$L = \lim_{u\to 0} \frac{\tan^{-1}\left(\frac{u}{1+\sqrt{1-u^2}}\right) - \frac{u}{2}}{u^3}$$

We can further simplify the inner expression by applying a trigonometric substitution. For $|u| < 1$, let $u = \sin(\theta)$, which implies $\theta = \sin^{-1}(u)$ and $\sqrt{1-u^2} = \cos(\theta)$. The argument becomes:

$$\frac{u}{1+\sqrt{1-u^2}} = \frac{\sin(\theta)}{1+\cos(\theta)}$$

Using the half-angle trigonometric identities $\sin(\theta) = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ and $1+\cos(\theta) = 2\cos^2\left(\frac{\theta}{2}\right)$:

$$\frac{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\cos^2\left(\frac{\theta}{2}\right)} = \tan\left(\frac{\theta}{2}\right)$$

Taking the inverse tangent of this simplified expression:

$$\tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right) = \frac{\theta}{2} = \frac{1}{2}\sin^{-1}(u)$$

Substituting this remarkable trigonometric identity back into our limit expression:

$$L = \lim_{u\to 0} \frac{\frac{1}{2}\sin^{-1}(u) - \frac{u}{2}}{u^3} = \frac{1}{2}\lim_{u\to 0} \frac{\sin^{-1}(u) - u}{u^3}$$

We now expand $\sin^{-1}(u)$ using its standard Maclaurin power series around $u = 0$:

$$\sin^{-1}(u) = u + \frac{1}{6}u^3 + \frac{3}{40}u^5 + O(u^7)$$

Subtracting the linear term u from the expansion gives:

$$\sin^{-1}(u) - u = \frac{1}{6}u^3 + O(u^5)$$

Substituting this algebraic approximation into the numerator:

$$L = \frac{1}{2}\lim_{u\to 0} \frac{\frac{1}{6}u^3 + O(u^5)}{u^3}$$

Dividing both numerator and denominator by the common factor u^3:

$$L = \frac{1}{2}\lim_{u\to 0}\left(\frac{1}{6} + O(u^2)\right) = \frac{1}{2}\cdot\frac{1}{6} = \frac{1}{12}$$

Sunday 10 May 2026

10.1 Daily Limit

Limit of an Inverse Square Sum Involving Quartics

Evaluate the limit:

$$\lim_{n\to\infty}\left(\sum_{k=1}^{n}\left[\frac{n}{\pi\sqrt{n^4+k^4}}-\frac{2nk^2}{\pi(n^2+k^2)\sqrt{n^4+k^4}}\right]\right)^{-2}$$

Solution: Let us first combine and simplify the general summand a_k inside the summation:

$$a_k=\frac{n}{\pi\sqrt{n^4+k^4}}\left[1-\frac{2k^2}{n^2+k^2}\right]=\frac{n}{\pi\sqrt{n^4+k^4}}\left[\frac{(n^2+k^2)-2k^2}{n^2+k^2}\right]=\frac{n(n^2-k^2)}{\pi(n^2+k^2)\sqrt{n^4+k^4}}$$

To express this summation as a Riemann sum, we factor out highest powers of n from each term. Dividing numerator and denominator by n^4:

$$a_k=\frac{1}{n}\cdot\frac{1-\left(\frac{k}{n}\right)^2}{\pi\left[1+\left(\frac{k}{n}\right)^2\right]\sqrt{1+\left(\frac{k}{n}\right)^4}}$$

Letting $\Delta x=\frac{1}{n}$ and $x_k=\frac{k}{n}$, the infinite sum converges to a definite integral S over $[0,1]$:

$$S=\lim_{n\to\infty}\sum_{k=1}^{n}a_k=\frac{1}{\pi}\int_0^1\frac{1-x^2}{(1+x^2)\sqrt{1+x^4}}\,dx$$

To evaluate this integral algebraically, we divide both the numerator and denominator of the integrand by x^2:

$$S=\frac{1}{\pi}\int_0^1\frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}\,dx$$

Notice the algebraic identity under the radical: $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2$. We apply the substitution $u=x+\frac{1}{x}$, which gives the differential $du=\left(1-\frac{1}{x^2}\right)dx=-\left(\frac{1}{x^2}-1\right)dx$: 1. As $x\to0^+$, $u\to\infty$. 2. As $x=1$, $u=1+1=2$.

Substituting these into our integral reverses the bounds:

$$S=\frac{1}{\pi}\int_\infty^2\frac{-du}{u\sqrt{u^2-2}}=\frac{1}{\pi}\int_2^\infty\frac{du}{u\sqrt{u^2-2}}$$

We evaluate this standard inverse secant integral using the trigonometric substitution $u=\sqrt{2}\sec(\theta)$, where $du=\sqrt{2}\sec(\theta)\tan(\theta)d\theta$ and $\sqrt{u^2-2}=\sqrt{2}\tan(\theta)$: 1. When $u=2$, $\sec(\theta)=\sqrt{2}\implies\theta=\frac{\pi}{4}$. 2. When $u\to\infty$, $\theta\to\frac{\pi}{2}$.

$$S=\frac{1}{\pi}\int_{\pi/4}^{\pi/2}\frac{\sqrt{2}\sec(\theta)\tan(\theta)}{\sqrt{2}\sec(\theta)\cdot\sqrt{2}\tan(\theta)}\,d\theta=\frac{1}{\pi\sqrt{2}}\int_{\pi/4}^{\pi/2}d\theta=\frac{1}{\pi\sqrt{2}}\left(\frac{\pi}{2}-\frac{\pi}{4}\right)=\frac{1}{4\sqrt{2}}$$

The original problem asks for the value of $L=S^{-2}$:

$$L=\left(\frac{1}{4\sqrt{2}}\right)^{-2}=(4\sqrt{2})^2=16\cdot2=32$$

Monday 11 May 2026

11.1 Daily Limit

Logarithmic Limit Involving Factorials and Exponents

Evaluate the limit:

$$\lim_{n\to\infty} \log_n \left(\ln \left(\frac{(n^2)!}{(n!)^n} \right) \right)$$

Solution: Let us define the argument of the outer base-n logarithm as A_n:

$$A_n = \ln \left(\frac{(n^2)!}{(n!)^n} \right) = \ln((n^2)!) - n\ln(n!)$$

To determine the asymptotic behavior of factorials for large n, we apply Stirling's Approximation Formula:

$$\ln(k!) = k\ln(k) - k + \frac{1}{2}\ln(2\pi k) + O\left(\frac{1}{k}\right)$$

We apply this asymptotic expansion individually to both terms in our expression: 1. For the first term where $k = n^2$:

$$\ln((n^2)!) = n^2\ln(n^2) - n^2 + \frac{1}{2}\ln(2\pi n^2) + O\left(\frac{1}{n^2}\right) = 2n^2\ln(n) - n^2 + \ln(n) + \frac{1}{2}\ln(2\pi) + O\left(\frac{1}{n^2}\right)$$

2. For the second term where $k = n$:

$$n\ln(n!) = n\left[n\ln(n) - n + \frac{1}{2}\ln(2\pi n) + O\left(\frac{1}{n}\right)\right] = n^2\ln(n) - n^2 + \frac{n}{2}\ln(n) + \frac{n}{2}\ln(2\pi) + O(1)$$

Subtracting the second expanded term from the first:

$$A_n = \left(2n^2\ln(n) - n^2\right) - \left(n^2\ln(n) - n^2\right) - \frac{n}{2}\ln(n) - \frac{n}{2}\ln(2\pi) + O(\ln n)$$

$$A_n = n^2\ln(n) - \frac{n}{2}\ln(n) - \frac{n}{2}\ln(2\pi) + O(\ln n)$$

We observe that for large n, the expression is strictly dominated by its leading asymptotic term:

$$A_n \sim n^2\ln(n) \implies A_n = n^2\ln(n)\left[1 - O\left(\frac{1}{n}\right)\right]$$

We now evaluate the original limit by changing the base-n logarithm to the natural logarithm:

$$L = \lim_{n\to\infty} \log_n(A_n) = \lim_{n\to\infty} \frac{\ln(A_n)}{\ln(n)}$$

Substituting our asymptotic factorization of A_n into the numerator:

$$\ln(A_n) = \ln\left(n^2\ln(n)\left[1 - O\left(\frac{1}{n}\right)\right]\right) = \ln(n^2) + \ln(\ln(n)) + \ln\left(1 - O\left(\frac{1}{n}\right)\right)$$

$$= 2\ln(n) + \ln(\ln(n)) + o(1)$$

Substituting this expanded numerator back into the fractional limit:

$$L = \lim_{n\to\infty} \frac{2\ln(n) + \ln(\ln(n)) + o(1)}{\ln(n)}$$

Dividing each individual term in the numerator by $\ln(n)$:

$$L = \lim_{n\to\infty} \left[2 + \frac{\ln(\ln(n))}{\ln(n)} + o\left(\frac{1}{\ln(n)}\right) \right]$$

Since logarithmic growth is strictly sub-polynomial, $\lim_{n\to\infty} \frac{\ln(\ln(n))}{\ln(n)} = 0$. Therefore:

$$L = 2 + 0 + 0 = 2$$